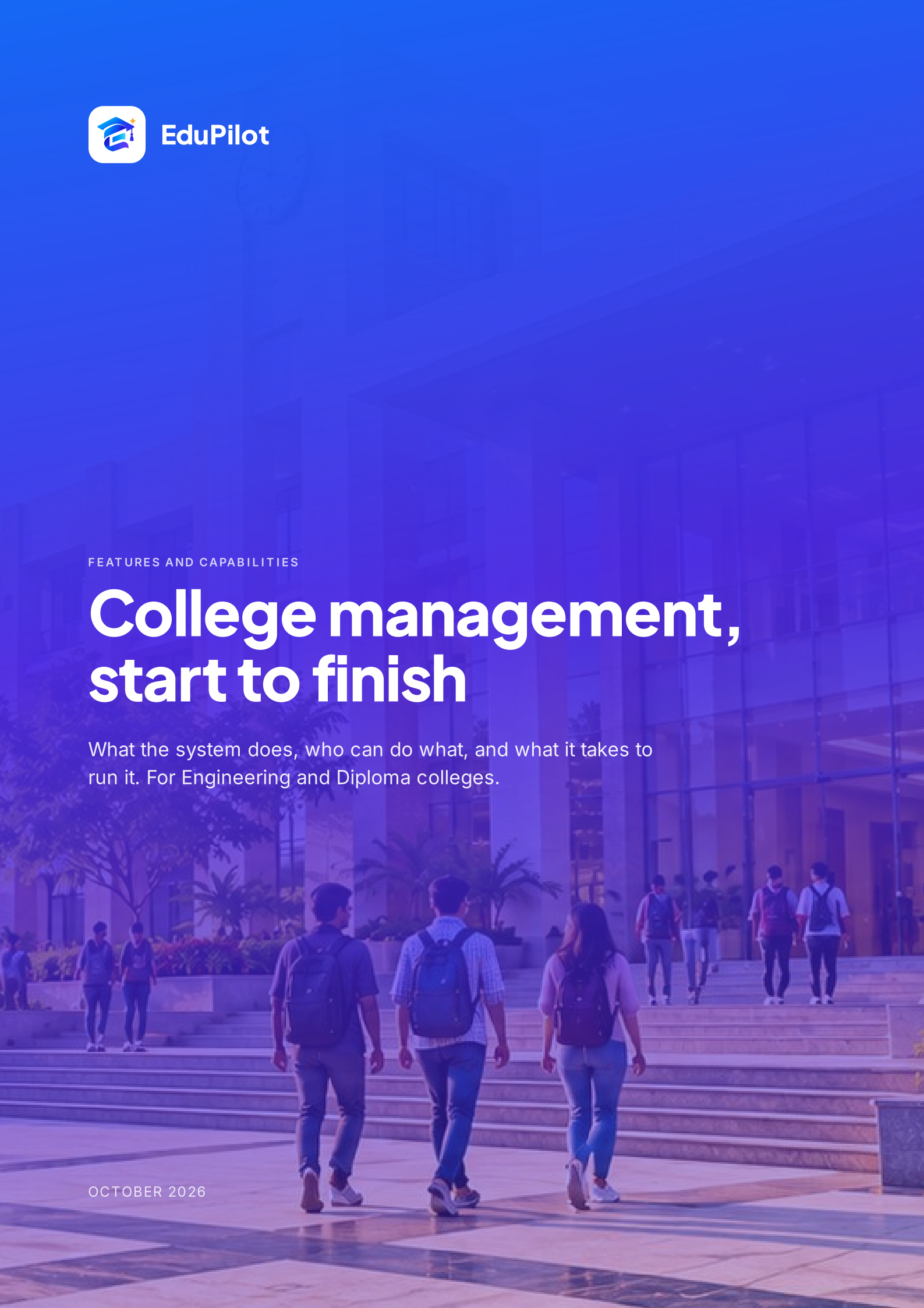
EduPilot

FEATURES AND CAPABILITIES

College management, start to finish

What the system does, who can do what, and what it takes to run it. For Engineering and Diploma colleges.



OCTOBER 2026

OVERVIEW

EduPilot at a glance

One web application for the academic, administrative and financial work of an Engineering or Diploma college, used by everyone from the principal to the parent.

7
roles, each with its own dashboard and menu

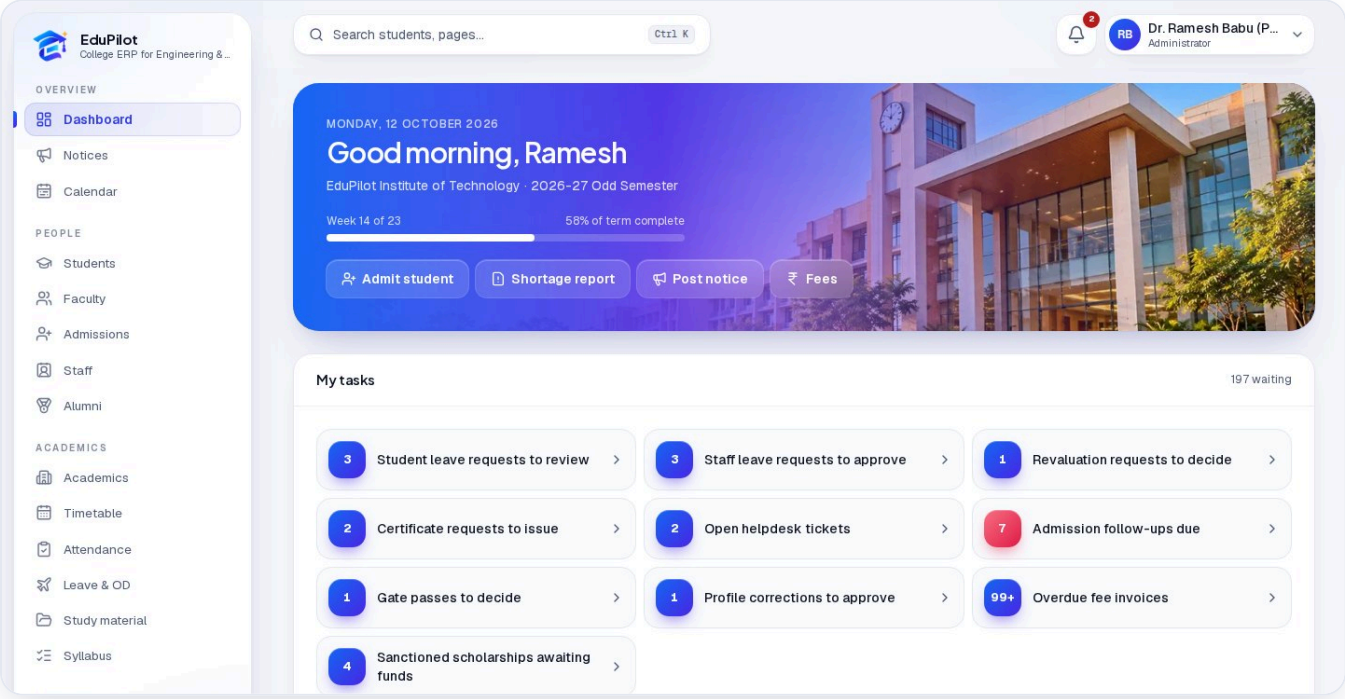
37
modules on the menu

96
screens

3
languages: English, Hindi, Telugu

92
automated tests passing (70 rule checks, 22 full user journeys)

200
students at once in the load test, with no failed page



The principal's dashboard: tasks waiting, then the institution's figures.

BUILT FOR

- B.Tech, M.Tech and Diploma (Polytechnic) programmes
- Semesters, lateral entry, internal and end-semester marks
- Attendance condonation, backlogs, supplementary exams
- AICTE, NAAC and NBA reporting needs

THE IDEA

- Each person sees only what their role allows
- Every dashboard starts with what is waiting on you
- The college sets its own rules without a programmer
- Anything printed carries the college's letterhead

OVERVIEW

Everything on the menu

37 modules in 7 groups. Each person's menu shows only the ones their role can use.

Overview

Dashboard	Home, KPIs and insights
Notices	Circulars and announcements
Calendar	Holidays, exams and events

People

Students	Admissions, profiles, import/export
Faculty	Staff profiles and workload
Admissions	Enquiry pipeline and follow-ups
Staff	Staff attendance, leave, appraisal, swaps
Alumni	Graduates and where they are now

Academics

Academics	Departments, programmes, curriculum
Timetable	Weekly class schedule
Attendance	Mark and track attendance
Leave & OD	Leave and on-duty requests
Study material	Notes and assignments
Syllabus	Lesson plan and coverage

Examinations

Marks entry	Assessments and marks
Exam schedule	Dates, rooms and seating
Hall tickets	Eligibility and admit cards
Results	SGPA, CGPA and grade cards
Revaluation	Re-check requests and grace marks
Supplementary	Backlog exam registration
Electives	Elective selection
Outcomes (CO-PO)	Attainment for accreditation
Feedback	Course feedback

Campus

Fees	Invoices, collections, receipts
Scholarships	Scheme claims and status
Placements	Drives, applications, offers
Library	Books, issues and returns
Hostel	Rooms and allocation
Transport	Routes, stops and passes
Inventory	Lab equipment and assets

Services

Gate pass	Visitors and student gate passes
Certificates	Bonafide, study, transfer
Helpdesk	Grievances and support
ID card	Digital identity card

System

Reports	Report builder, accreditation, schedules
Administration	Users, branding, audit log
My Profile	Photo, language and password

WHAT IT DOES

Features, area by area

01 Academic set-up

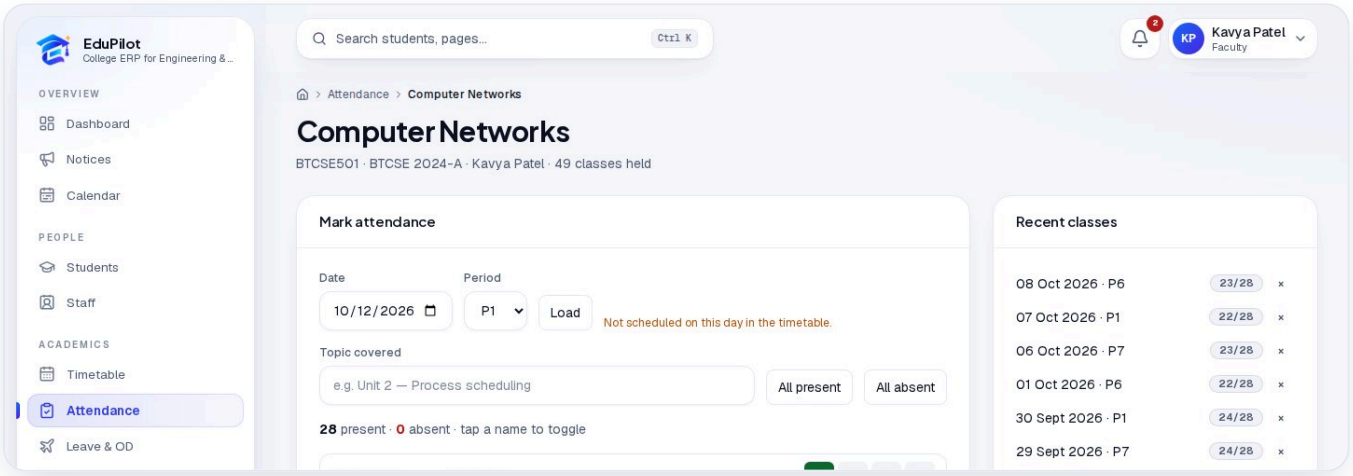
The structure everything else depends on.

- Departments, programmes (Diploma, B.Tech, M.Tech), batches and sections
- Curriculum with L-T-P hours, credits and the internal / external marks split
- Course allocation with teacher workload; this term's courses generated from the curriculum
- Drag-and-drop timetable that refuses teacher clashes
- Academic calendar; holidays and Sundays block attendance
- Semester promotion with a review of who is held back; graduation creates alumni

02 Daily teaching

What teachers and students do every day.

- Period-wise attendance in a few taps: present, absent, late, on-duty
- Works on a phone, by swipe, and without a connection
- Attendance rule with a condonation band, set by the college
- Shortage report with email alerts and WhatsApp reminders to parents
- Student leave and on-duty requests that correct attendance when approved
- Study material, online assignments, grading with feedback
- Lesson plans with syllabus coverage visible to the HOD
- Substitute teachers and class swaps between colleagues



Taking attendance: everyone starts present; tap the absentees.

WHAT IT DOES

Features, continued

03 Examinations and results

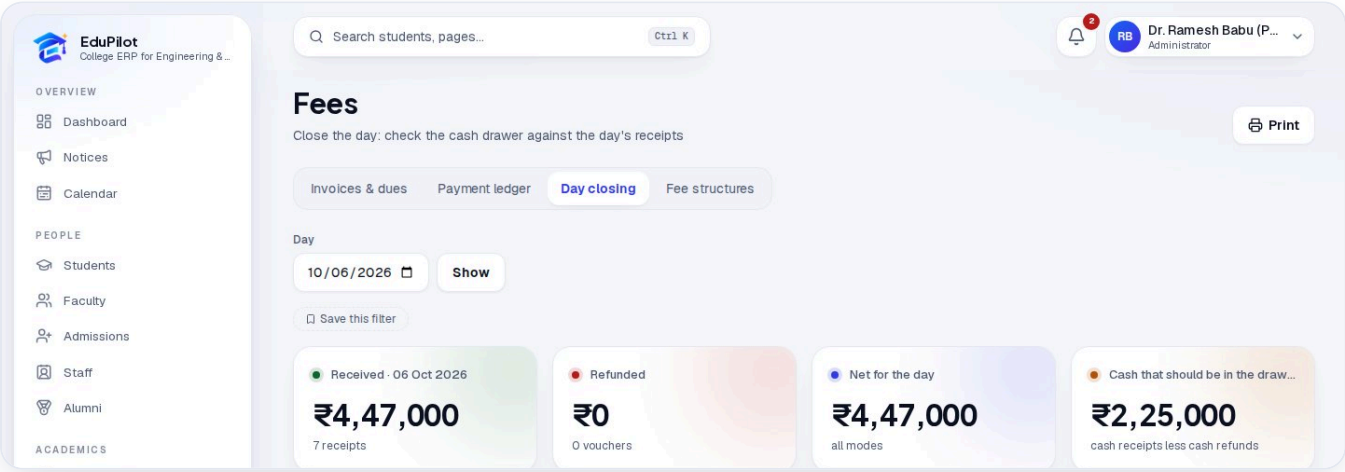
From internal marks to the grade card.

- Assessment plans; spreadsheet-style marks grid with paste from Excel
- Mid-term and end-semester schedule with clash checks
- Exam halls, generated seating, invigilation duty shared evenly
- Hall tickets only when attendance and fees are in order; whole batch as one PDF
- Grades, SGPA, CGPA and backlogs computed on publishing
- The college's own pass marks and grade scale
- Revaluation requests and grace marks, both recorded
- Supplementary registration with automatic fee; seat-limited electives
- Outcome attainment (CO-PO) for NBA / NAAC; anonymous course feedback

04 Fees and finance

Every rupee accounted for.

- Fee structures per programme and year; invoices for a whole batch in one click
- Instalment plans, per-day late fines with a cap, concessions
- Counter collection by cash, UPI, card, bank transfer or DD
- Online payment by students and parents (Razorpay, with a built-in test checkout)
- Numbered receipts with the amount in words, on the college letterhead
- Refunds against a receipt, with their own voucher
- Day book and end-of-day cash closing against the books
- Scholarship claims that credit the invoice when received
- Fee statement that becomes a no-dues certificate when nothing is owed



Day closing: the day's receipts by mode, against the cash counted.

WHAT IT DOES

Features, continued

05 Admissions and student records

From first enquiry to alumni.

- Admissions board: enquiry, applied, documents verified, admitted; follow-up dates
- Admission form and bulk import from a spreadsheet; lateral entry handled
- One profile per student: attendance, results, fees, placements
- ID cards with photo and QR code, one at a time or a whole batch
- Parent logins with a read-only view of their child
- Parent contact log: every reminder, call and meeting
- Contact corrections requested by students, approved by the office
- Alumni records created on graduation

06 Student services

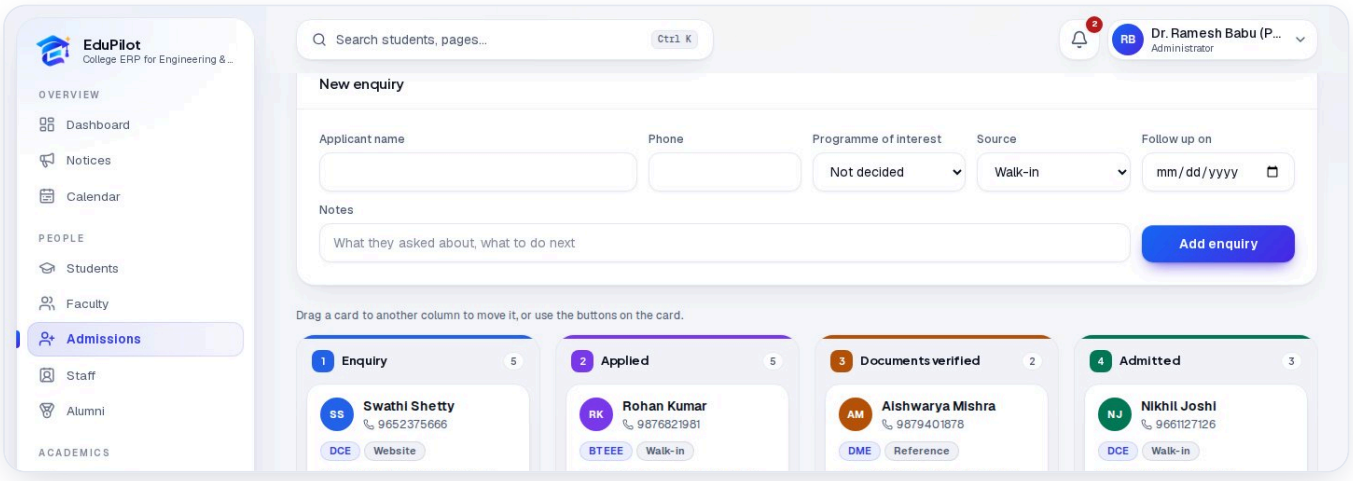
Requests handled without a queue at the office.

- Bonafide, study, conduct and transfer certificates with serial numbers
- One certificate issued to a whole batch for a scholarship round
- Helpdesk tickets with threaded replies
- Notices to everyone, students or staff; a personal notification inbox
- Gate passes: request, approve, out, returned, with late returns flagged
- Placement drives with automatic eligibility, rounds and offer letters

07 Campus operations

The facilities around the classroom.

- Library catalogue, issue and return, overdue fines
- Hostels, rooms, beds and allotment
- Transport routes, stops with times, bus passes
- Inventory of lab equipment and assets, with issue and return
- Visitor sign-in and sign-out



The admissions board: drag a card to the next stage.

WHAT IT DOES

Features, continued

08 Staff and payroll

The people who run the college.

- Faculty profiles with teaching history
- Daily staff attendance; leave with yearly balances and approval
- Payroll: salaries, a monthly run from attendance, printable payslips
- Appraisal records: publications, FDPs, patents, projects, awards
- A front-office role with no access to marks, money or settings

09 Reports and accreditation

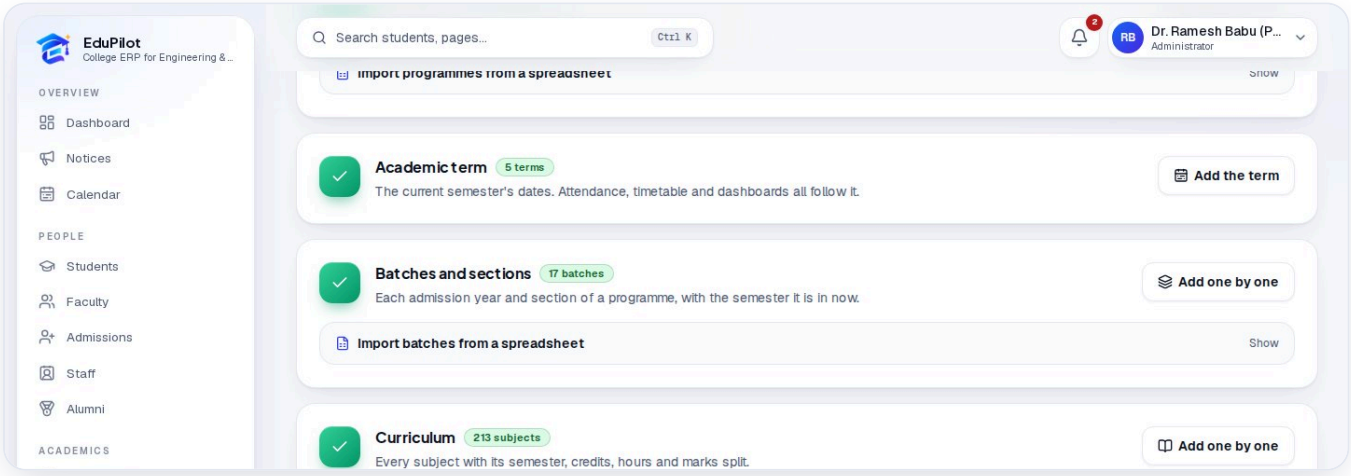
Getting information out.

- Role-specific dashboards with charts that expand and download
- Report builder: seven reports, chosen columns, Excel (CSV) or PDF
- Accreditation data sheets for AICTE, NAAC and NBA files
- Weekly or monthly summaries by email
- Year-end archive of every record

10 Administration

Keeping the system in order.

- Guided college setup with spreadsheet import and templates
- Branding: name, logo, colour, letterhead on every document
- Rules page: attendance, pass marks, grade scale, promotion, placements
- Users, roles, disabling accounts, password resets
- Audit log of every change and sign-in
- System health: backups, server errors, email, storage



College setup: a checklist with spreadsheet import for each bulk step.

ACCESS

Who can use what

A dot means the module is on that role's menu. Inside a module, access narrows further: an HOD to one department, a teacher to their own courses, a student or parent to one record.

Module	Principal / Admin	HOD	Faculty	Accounts	Office	Student	Parent
OVERVIEW							
Dashboard	●	●	●	●	●	●	●
Notices	●	●	●	●	●	●	●
Calendar	●	●	●	●	●	●	●
PEOPLE							
Students	●	●	●	●	●		
Faculty	●	●					
Admissions	●				●		
Staff	●	●	●	●	●		
Alumni	●	●					
ACADEMICS							
Academics	●	●					
Timetable	●	●	●	●	●	●	●
Attendance	●	●	●			●	●
Leave & OD	●	●	●			●	
Study material	●	●	●			●	
Syllabus	●	●	●				
EXAMINATIONS							
Marks entry	●	●	●				
Exam schedule	●	●	●			●	●
Hall tickets	●	●				●	
Results	●	●				●	●
Revaluation	●	●				●	
Supplementary	●					●	
Electives	●	●				●	
Outcomes (CO-PO)	●	●	●				
Feedback	●	●	●			●	
CAMPUS							
Fees	●			●		●	●
Scholarships	●			●			
Placements	●	●				●	
Library	●		●		●	●	
Hostel	●				●	●	
Transport	●				●	●	
Inventory	●	●			●		
SERVICES							
Gate pass	●				●	●	
Certificates	●				●	●	
Helpdesk	●	●	●	●	●	●	●
ID card						●	
SYSTEM							
Reports	●	●		●			
Administration	●						
My Profile	●	●	●	●	●	●	●

Parents see their child's records read-only and can pay fees. Office staff are refused fees, marks, results, reports and administration even if they type the address.

PLATFORM

How it is built to be trusted

Beyond features: how the system protects data, stays fast, and keeps running.

Security

- Every page and action checks the person's role on the server
- HODs limited to their department, teachers to their courses, students and parents to one record
- Passwords stored hashed; lockout after five wrong attempts
- Optional two-step sign-in for staff, by emailed code
- Changing a password signs out other devices

Speed

- Dashboards show the banner and tasks first; figures follow
- Totals and percentages are worked out by the database
- Tested with 200 students at once: 3,023 page loads, none failed
- About 95 page loads a second on a laptop; a server will do more

Reliability

- Automatic nightly backup, newest 14 kept, optional second copy elsewhere
- One-command restore, also used to move servers
- Server errors logged, counted and emailed to the administrator
- The app and database restart by themselves after a reboot

On a phone

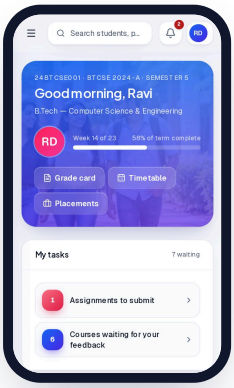
- Runs in the phone's browser; can be added to the home screen
- Bottom tab bar for students, parents and teachers
- Tables become cards with the heading beside each value
- Attendance can be marked without a connection

Languages and access

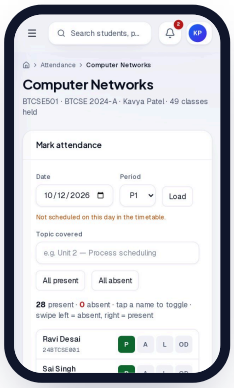
- Menu, page titles and common labels in English, Hindi and Telugu
- Adjustable text size; keyboard shortcuts; visible focus outlines
- Screens scanned for accessibility problems in all seven roles
- Translations should be reviewed by a native speaker before launch

Documents and data

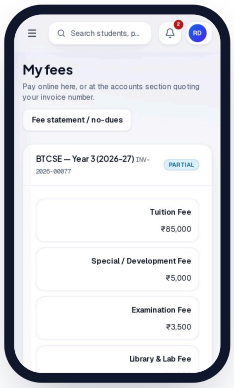
- Receipts, hall tickets, certificates, payslips and reports as PDF or print
- Every list page prints cleanly with the college name and date
- Spreadsheet import for setup and students; CSV export from lists and reports
- Saved filters on list pages; search across students and pages



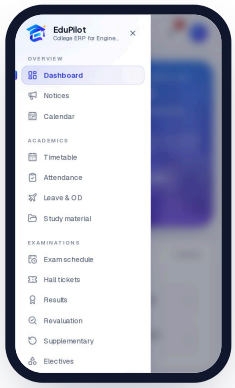
Student dashboard



Teacher's attendance



Fees and Pay now



Full menu

ADOPTION

Going live, and what is not included

What the college provides, how the product was checked, and where its edges are.

WHAT IS NEEDED TO GO LIVE

A server or cloud machine
The app and its MySQL database run in Docker.

A web address with HTTPS
For example erp.yourcollege.edu.

A mail account (optional)
Any SMTP account. Without it, emails are stored and not delivered, and two-step sign-in cannot be switched on.

A Razorpay account (optional)
For real online payments. This path follows Razorpay's documented flow and should be tried with the college's own account before going live.

The college's data
Imported from spreadsheets through the setup checklist.

NOT INCLUDED

- SMS and automatic WhatsApp messages. WhatsApp reminders open the staff member's own WhatsApp with the message written.
- An app-store mobile app. It runs in the phone's browser.
- Official accreditation templates. The accreditation pages give the figures to copy into them.
- Income tax in payroll. Payslips are a salary statement.
- Relative grading. Grades follow a fixed scale the college sets.

HOW IT WAS CHECKED

70
checks of the rules: grading, pass marks, fee status, payroll, eligibility, passwords

22
full journeys run in a real browser, across every role, on every change

7 roles
every screen opened as each role on desktop, and at phone size for four of them, looking for errors and layout breaks

376
invoices used to confirm the faster fee calculation agrees with the original rules exactly

Honest limits of that testing. It was done on a demonstration college of 424 students, 35 staff, 5 departments and 8 programmes, on a laptop. Email delivery and Razorpay payments have not been run against real accounts. The Hindi and Telugu text has not been reviewed by a native speaker. A pilot with one department's real data is the right next step.

A screen-by-screen walk-through of every feature and workflow is in the companion document, **EduPilot Product Guide**.